Lost in Retrieval: When Language Models Fail to Use Available Evidence in RAG Systems

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Abstract

Retrieval-augmented generation systems are often evaluated under the assumption that retrieving more documents improves generation by increasing the chance of including relevant evidence. However, larger retrieval contexts can also make it harder for the generator to decide whether to answer or abstain, especially when instructed to rely only on retrieved evidence. In this extended abstract, we investigate this behavior as a knowledge reconciliation phenomenon between contextual and parametric knowledge. We propose an evidence-aware taxonomy to distinguish justified abstentions, false abstentions, and unsupported answers, and we apply it to study how large language models reconcile knowledge across evidence-availability conditions. Our initial results show that, while retrieval depth improves evidence availability, it does not uniformly improve generator effectiveness.

Keywords

Retrieval-Augmented Generation, Large Language Models, Knowledge Reconciliation.

1. Introduction

Context. Retrieval-Augmented Generation (RAG) has become a fundamental paradigm for knowledge-intensive tasks, combining the parametric knowledge stored in Large Language Models (LLMs) with knowledge retrieved from human-validated external corpora [1, 2, 3, 4, 5]. In Question Answering (QA) tasks, this design is commonly motivated by the need to improve factuality and attribution by grounding answer generations in retrieved evidence [6, 7, 8]. At the same time, the generator must decide how to use the retrieved context, whether to trust it, ignore it, or complement it with its own parametric knowledge [9, 10, 11, 12]. This decision becomes more delicate as retrieval depth increases, since larger contexts may improve recall but also introduce irrelevant or distracting evidence [13, 14, 15, 16]. While RAG is often evaluated through retrieval effectiveness or final answer correctness, less attention has been paid to how generators decide whether the retrieved context is sufficient to answer.

Motivation. RAG systems are increasingly deployed in proprietary and domain-specific applications, where user queries depend on private documents that are not encoded in the parametric knowledge of LLMs [17]. In these settings, retrieval depth becomes a practical reliability aspect: retrieving more documents increases the chance of exposing the generator to the required evidence [18]. Yet, this coverage-oriented strategy can introduce additional noise, making it harder for the generator to determine whether the context is sufficient to answer. This tension is particularly important when LLMs are instructed to rely exclusively on retrieved documents and abstain otherwise, as a safeguard against hallucinations [19]. Under this constraint, abstention is not merely a fallback behavior, but a key signal of how the generator reconciles retrieved evidence with its own parametric knowledge.

Open Issues. Several lines of work analyzed this problem in RAG, but they typically addressed distinct aspects. Retrieval-oriented studies focus on retrieval quality, context selection, and retrieval-depth

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optimization, showing that better evidence retrieval is crucial for downstream generation [3, 4], but they often overlook whether the generator correctly recognizes when retrieved content is sufficient. Work on hallucination mitigation and faithfulness proposes strategies to reduce unsupported generation [7, 11, 12, 8], but mainly evaluates whether answers are supported rather than why models abstain. A third line studies unanswerability, abstention, and conflicts between contextual and parametric knowledge [20, 9, 10, 21]. Yet, abstention is rarely analyzed as an evidence-aware behavior: a refusal is justified when no relevant evidence is retrieved, but erroneous when such evidence is present and ignored. Without conditioning generator behavior on retrieval recall, it remains unclear whether failures originate from missing evidence, noisy contexts, or unexploited relevant documents.

Our Contributions. In this extended abstract, we conduct a preliminary analysis of knowledge reconciliation in RAG systems adopted for QA tasks, by studying when generators answer or abstain depending on the availability of correct supporting evidence. We propose an evidence-aware taxonomy that separates generation outcomes by evidence availability, distinguishing correct/incorrect answers, false/justified abstentions, parametric answers, and hallucinated answers. Finally, we analyze how LLM generators reconcile contextual and parametric knowledge across these evidence-availability conditions.

2. Methodology

In this section, we formalize our RAG-based QA setting and define the evidence-aware taxonomy used to study when generators answer, abstain, or rely on unsupported knowledge. To support our taxonomy, we define evidence availability at retrieval depth k with $\rho_i^{(k)}$. Specifically, $\rho_i^{(k)} = 1$ indicates that at least one answer-supporting document is present in the retrieved context for a given question q_i, while $\rho_i^{(k)} = 0$ indicates that no supporting evidence is available to the generator.

2.1. Evidence-aware Generation Taxonomy Definition

We classify generated outputs according to three observable properties: whether answer-supporting evidence is available in the retrieved context ($\rho_i^{(k)}$), whether the output matches a gold answer through a correctness function $\mathrm{corr}(y_i^{(k)})$, and whether the model abstains ($y_i^{(k)} = \perp$).

When relevant evidence is present in the retrieved context ($\rho_i^{(k)} = 1$), we distinguish three outcomes:

- Correct Answer ($\rho_i^{(k)} = 1 \wedge y_i^{(k)} \neq \perp \wedge \mathrm{corr}(y_i^{(k)}) = 1$): The model produces a correct answer.
- Incorrect Answer ($\rho_i^{(k)} = 1 \wedge y_i^{(k)} \neq \perp \wedge \mathrm{corr}(y_i^{(k)}) = 0$): The model produces an incorrect answer despite having access to relevant evidence.
- False Abstention ($\rho_i^{(k)} = 1 \wedge y_i^{(k)} = \perp$): The model abstains, though there is relevant evidence.

When no relevant evidence is retrieved ($\rho_i^{(k)} = 0$), we distinguish three outcomes:

- Justified Abstention ($\rho_i^{(k)} = 0 \wedge y_i^{(k)} = \perp$): The model correctly abstains.
- Parametric Answer ($\rho_i^{(k)} = 0 \wedge y_i^{(k)} \neq \perp \wedge \mathrm{corr}(y_i^{(k)}) = 1$): The model produces a correct answer despite the absence of relevant evidence (relying on internal knowledge).
- Hallucinated Answer ($\rho_i^{(k)} = 0 \wedge y_i^{(k)} \neq \perp \wedge \mathrm{corr}(y_i^{(k)}) = 0$): The model produces an incorrect answer without relevant evidence.

2.2. Taxonomy-aware RAG Operationalization

We operationalize the generator phase of the RAG system under three methodological constraints. First, we use a minimal two-stage RAG architecture consisting of retrieval followed by generation, avoiding re-ranking or query rewriting to observe how the generator behaves when exposed to different amounts of retrieved evidence. Second, we use an abstention-aware prompt instructing the model to answer

Table 1
Evidence-aware distribution of generation outcomes across retrieval depths (k). Values are percentages over all queries, separating evidence-present and evidence-absent cases. Best values per column are in bold.

k	Recall (%)	Evidence Present ($\rho_i^{(k)} = 1$)			Evidence Absent ($\rho_i^{(k)} = 0$)		
		Correct Answer (% ↑)	Incorrect Answer (% ↓)	False Abstention (% ↓)	Justified Abstention (% ↑)	Parametric Answer (% ↓)	Hallucinated Answer (% ↓)
Llama-3.1 8B							
1	62.3	52.7	**3.0**	**6.7**	**22.2**	3.1	12.3
3	74.4	58.3	4.5	11.6	15.3	0.8	9.5
5	78.2	61.8	5.6	10.8	11.5	0.7	9.6
10	82.0	64.0	7.1	10.9	8.7	**0.5**	8.8
15	83.8	66.0	7.6	10.1	6.8	0.6	8.8
20	84.8	**66.2**	7.5	11.0	6.5	**0.5**	**8.3**
Mistral 7B							
1	62.3	53.4	**3.8**	5.1	**18.6**	3.2	15.8
3	74.4	62.2	6.1	6.1	11.8	1.3	12.6
5	78.2	65.4	7.7	5.2	8.2	0.9	12.6
10	82.0	68.4	9.5	4.1	5.0	0.8	12.2
15	83.8	69.4	10.3	4.0	3.9	0.7	11.6
20	84.8	**70.1**	11.0	**3.7**	3.2	**0.6**	**11.4**

only when the context contains sufficient evidence, and to output an "unanswerable" response ($\perp$) otherwise [20]. Finally, answer correctness is evaluated using a cover-based criterion suited to factoid QA, checking if the normalized generated output contains at least one acceptable gold answer [4].

3. Preliminary Experiments

The goal of our preliminary experiments is to investigate how LLM generators reconcile contextual and parametric knowledge across different evidence-availability conditions. To this end, we instantiate the minimal RAG pipeline with FlashRAG [22], evaluating on the TriviaQA evaluation set (N = 11,313) [23]. We experiment with two instruction-tuned generators: `Llama-3.1 8B` [24] and `Mistral 7B` [25], under the same abstention-aware prompt and deterministic decoding (temperature 0). Retrieval uses E5 embeddings [26] with FAISS inner-product search across different retrieval depths (k).

Table 1 reports the outcome distribution across retrieval depths based on our taxonomy. The results show that increasing retrieval depth improves evidence availability, with recall rising from 62.3% at $k = 1$ to 84.8% at $k = 20$. Under the *evidence-present* condition, both models benefit from larger contexts, but in different ways: Mistral produces more correct answers and fewer false abstentions, while Llama is more conservative, with lower incorrect answers but higher false abstention rates as depth increases. Importantly, the presence of relevant evidence does not guarantee reliable generation, as incorrect answers also increase with retrieval depth. The *evidence-absent* branch highlights a different reliability issue. As k increases, justified abstentions decrease, while non-abstaining behavior persists even when no relevant evidence is retrieved. Parametric answers remain rare, suggesting that correct unsupported answers are not the dominant failure mode. Instead, hallucinated answers account for a substantial fraction of evidence-absent cases, especially for Mistral. Overall, the taxonomy shows that retrieval depth improves access to evidence, but does not by itself ensure contextual fidelity.

4. Conclusions and Future Work

This work presented a preliminary study of knowledge reconciliation in retrieval-augmented generation systems, focusing on how LLM generators behave depending on whether answer-supporting evidence is available in the retrieved context. To this end, we introduced an evidence-aware taxonomy that

distinguishes correct and incorrect answers, false and justified abstentions, parametric answers, and hallucinated answers. Experiments on TriviaQA with two instruction-tuned generators show that increasing retrieval depth substantially improves evidence availability, but does not translate uniformly into more reliable generation. In particular, relevant evidence may still be ignored or misused, while unsupported answers can persist even when no supporting evidence is retrieved. These results suggest that evaluating RAG systems only through retrieval recall or final answer correctness can obscure important differences in how generators reconcile contextual and parametric knowledge.

The current study also presents limitations that motivate our future work. First, our analysis is restricted to a single QA dataset and a limited set of generators and retrieval configurations. We therefore plan to extend the evaluation across additional datasets, domains, retrievers, and LLM families to assess whether the observed reconciliation patterns generalize beyond the current setting. Second, evidence availability is presently modeled as a binary condition, based on whether at least one answer-supporting document is present in the retrieved context. Future work will investigate finer-grained notions of evidence quality, including evidence completeness, redundancy, relevance strength, and position within the retrieved context, to better characterize when evidence is effectively usable by the generator. Third, the present experiments rely on a fixed abstention-aware prompting strategy and do not isolate the mechanisms responsible for the observed behaviors. We plan to study how prompting, context ordering, retrieval noise, and conflicts between contextual and parametric knowledge affect the balance between answering and abstaining. Finally, while our taxonomy characterizes observable outcomes, it does not yet explain why a model follows, ignores, or overrides retrieved evidence. A natural next step is therefore to investigate model-internal and behavioral signals that may help predict these reconciliation decisions and support more reliable evidence-aware generation.

Declaration on Generative AI

During the preparation of this work, the author(s) used GenAI tools in order to refine writing, document code, and edit the paper. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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